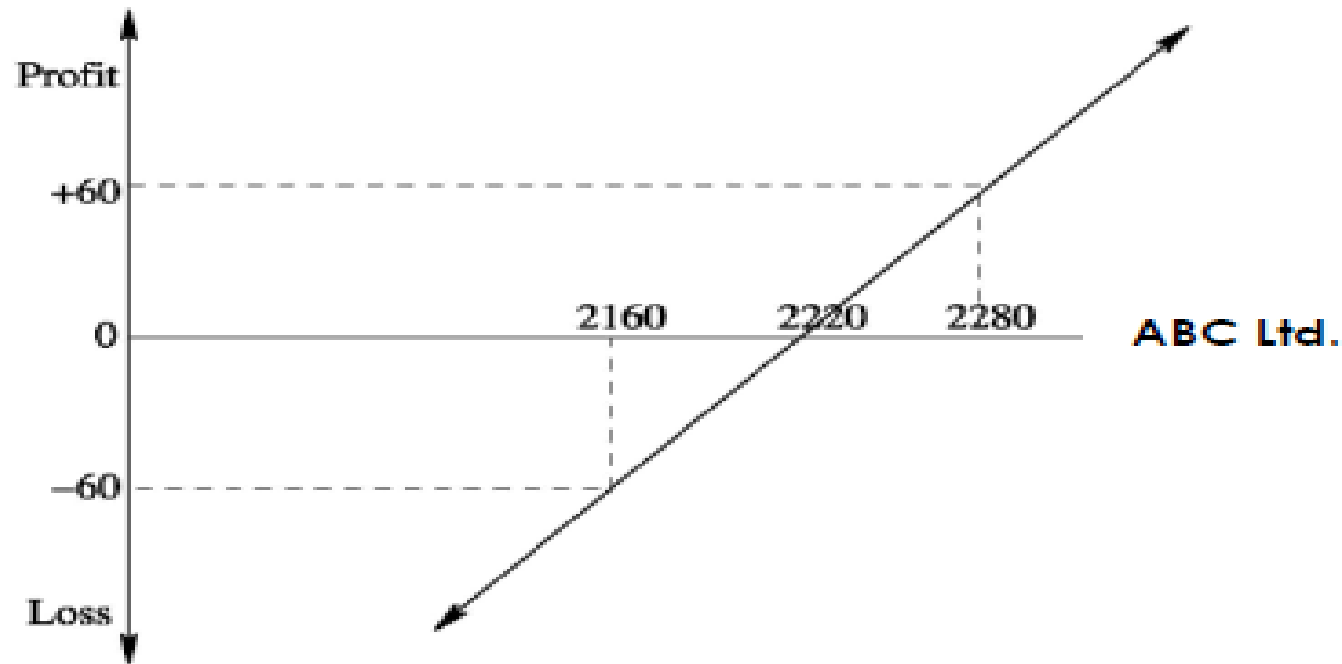


Option Trading Strategy

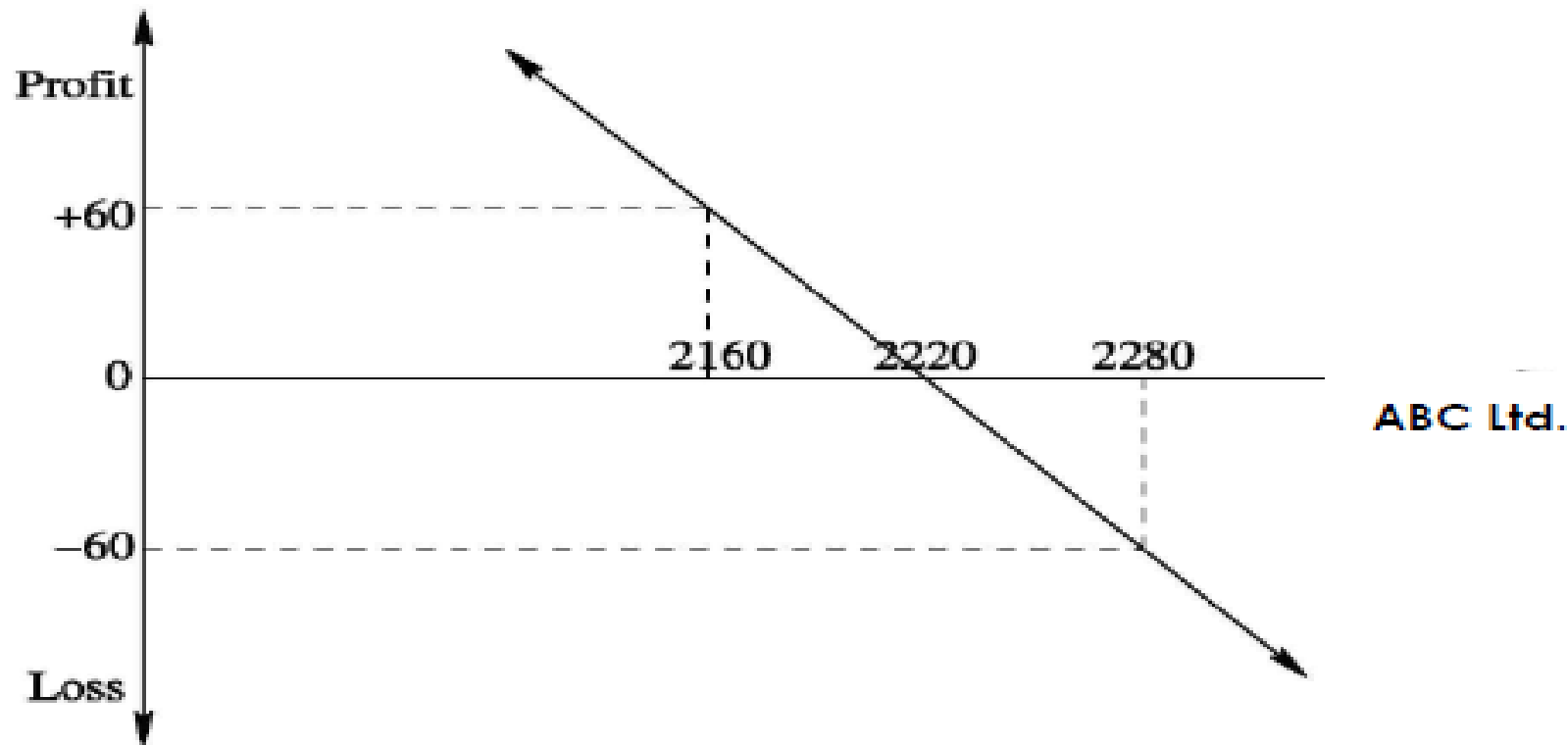
Payoff profile of buyer of asset: Long asset

- The figure shows the profits/losses from a long position on ABC Ltd.. The investor bought ABC Ltd. at Rs. 2220. If the share price goes up, he profits. If the share price falls, he loses.



Payoff profile for seller of asset: Short asset

- The figure shows the profits/losses from a short position on ABC Ltd.. The investor sold ABC Ltd. at Rs. 2220. If the share price falls, he profits. If the share price rises, he loses.



Long call

Buying a call is the most basic of all options strategies. It constitutes the first options trade for someone already familiar with buying / selling stocks and would now want to trade options. Buying a call is an easy strategy to understand. When you buy it means you are bullish. Buying a Call means you are very bullish and expect the underlying stock / index to rise in future.

When to Use: Investor is very **bullish** on the stock / index.

Risk: Limited to the Premium. (Maximum loss if market expires at or below the option strike price).

Reward: Unlimited

Breakeven: Strike Price + Premium

Example

Mr. XYZ is bullish on Nifty on 24th June, when the Nifty is at 4191.10. He buys a call option with a strike price of Rs. 4600 at a premium of Rs. 36.35, expiring on 31st July. If the Nifty goes above 4636.35, Mr. XYZ will make a net profit (after deducting the premium) on exercising the option. In case the Nifty stays at or falls below 4600, he can forego the option (it will expire worthless) with a maximum loss of the premium.

Strategy : Buy Call Option

	Current Nifty index	4191.10
Call Option	Strike Price (Rs.)	4600
Mr. XYZ Pays	Premium (Rs.)	36.35
	Break Even Point (Rs.) (Strike Price + Premium)	4636.35

The payoff schedule

On expiry Nifty closes at	Net Payoff from Call Option (Rs.)
4100.00	-36.35
4300.00	-36.35
4500.00	-36.35
4636.35	0
4700.00	63.65
4900.00	263.65
5100.00	463.65
5300.00	663.65

The payoff chart (Long Call)



ANALYSIS: This strategy limits the downside risk to the extent of premium paid by Mr. XYZ (Rs. 36.35). But the potential return is unlimited in case of rise in Nifty. A long call option is the simplest way to benefit if you believe that the market will make an upward move and is the most common choice among first time investors in Options. As the stock price / index rises the long Call moves into profit more and more quickly.

Short call

A Call option means an Option to buy. Buying a Call option means an investor expects the underlying price of a stock / index to rise in future. Selling a Call option is just the opposite of buying a Call option. Here the seller of the option feels the underlying price of a stock / index is set to fall in the future.

When to use: Investor is very aggressive and he is **very bearish** about the stock / index.

Risk: Unlimited

Reward: Limited to the amount of premium

Break-even Point: Strike Price + Premium

Example:

Mr. XYZ is bearish about Nifty and expects it to fall. He sells a Call option with a strike price of Rs. 2600 at a premium of Rs. 154, when the current Nifty is at 2694. If the Nifty stays at 2600 or below, the Call option will not be exercised by the buyer of the Call and Mr. XYZ can retain the entire premium of Rs. 154.

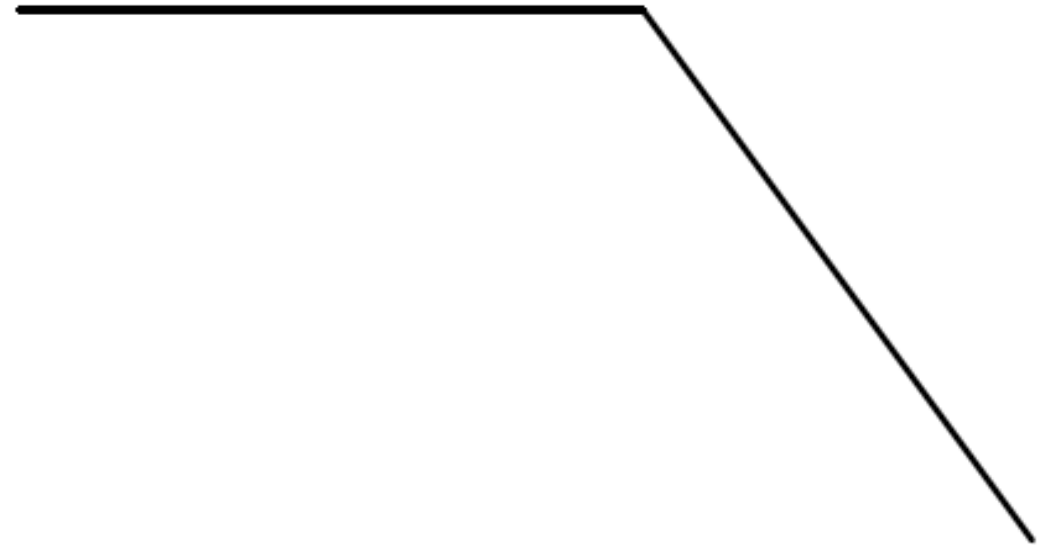
Strategy : Sell Call Option		
	Current Nifty index	2694
Call Option	Strike Price (Rs.)	2600
Mr. XYZ receives	Premium (Rs.)	154
	Break Even Point (Rs.) (Strike Price + Premium)*	2754

* Breakeven Point is from the point of Call Option Buyer.

The payoff schedule

On expiry Nifty closes at	Net Payoff from the Call Options (Rs.)
2400	154
2500	154
2600	154
2700	54
2754	0
2800	-46
2900	-146
3000	-246

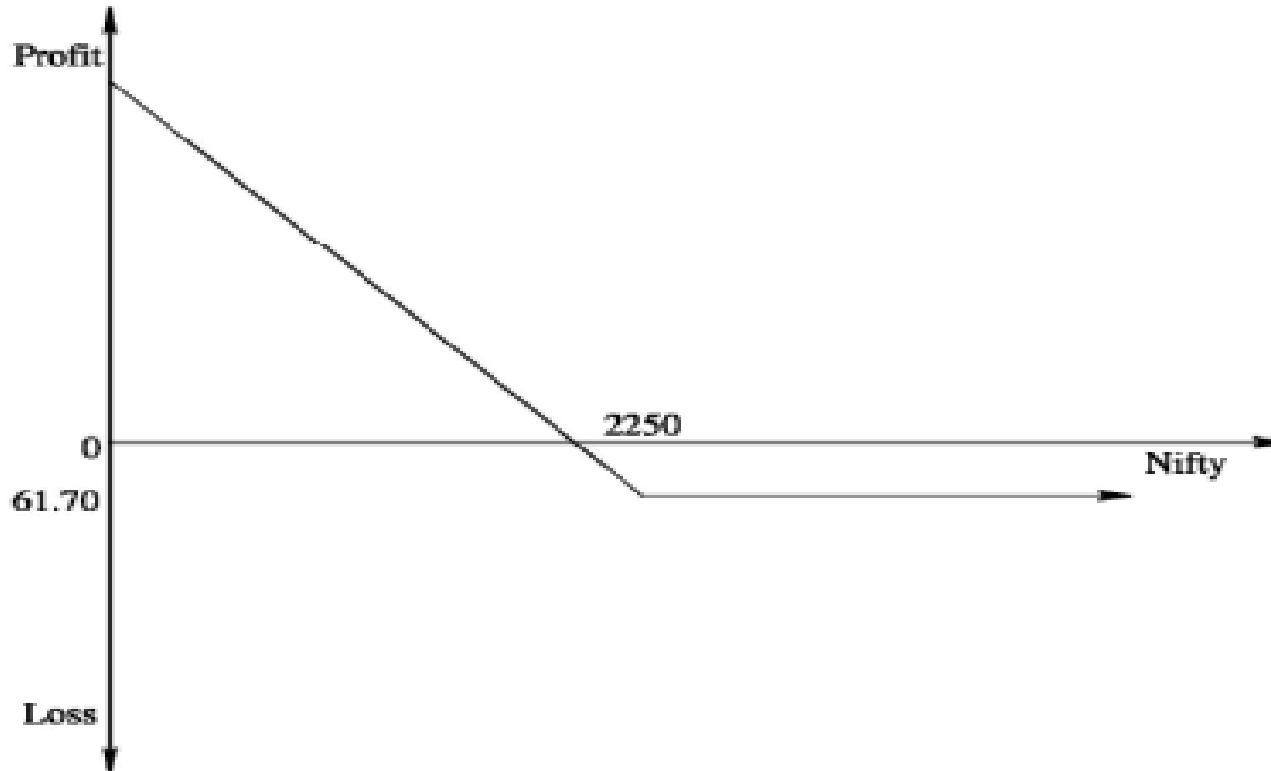
The payoff chart (Short Call)



ANALYSIS: This strategy is used when an investor is very aggressive and has a strong expectation of a price fall (and certainly not a price rise). This is a risky strategy since as the stock price / index rises, the short call loses money more and more quickly and losses can be significant if the stock price / index falls below the strike price. Since the investor does not own the underlying stock that he is shorting this strategy is also called Short Naked Call.

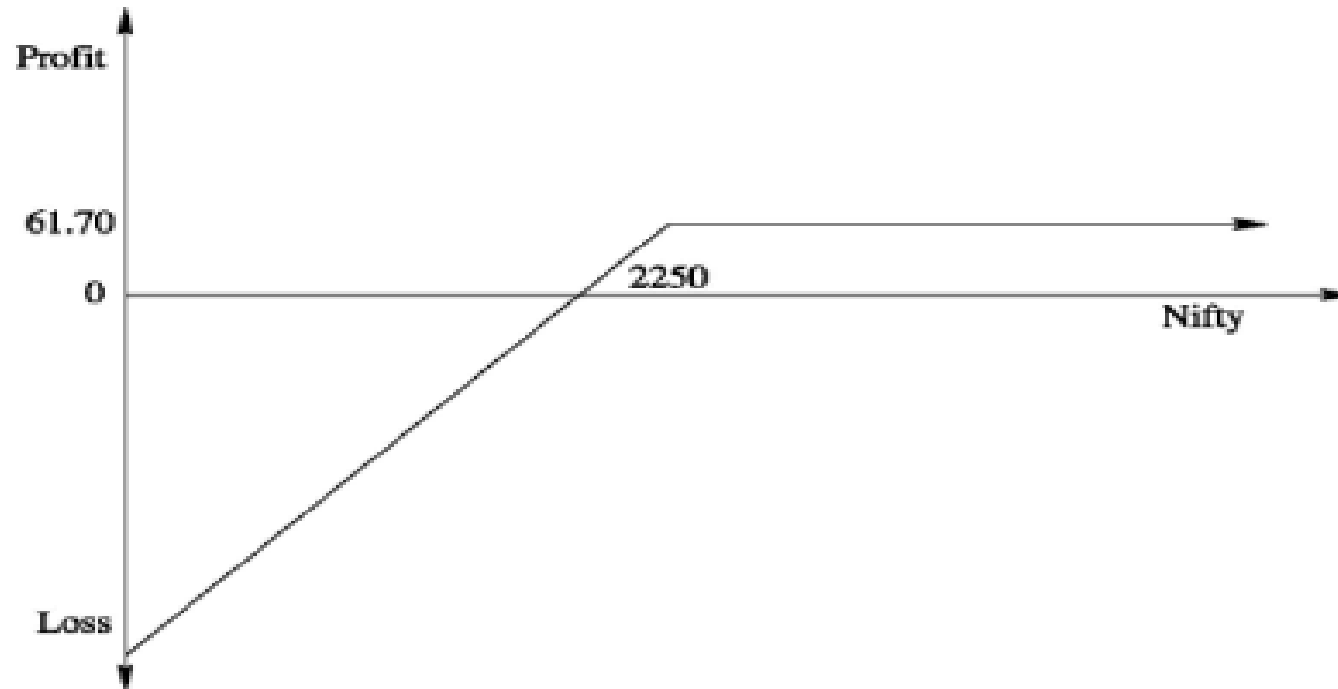
Payoff profile for buyer of put options: Long put

- The figure shows the profits/losses for the buyer of a three-month Nifty 2250 put option. As can be seen, as the spot Nifty falls, the put option is in-the-money. If upon expiration, Nifty closes below the strike of 2250, the buyer would exercise his option and profit to the extent of the difference between the strike price and Nifty-close. The profits possible on this option can be as high as the strike price. However if Nifty rises above the strike of 2250, he lets the option expire. His losses are limited to the extent of the premium he paid for buying the option.



Payoff profile for writer (seller) of put options: Short put

- The figure shows the profits/losses for the seller of a three-month Nifty 2250 put option. As the spot Nifty falls, the put option is in-the-money and the writer starts making losses. If upon expiration, Nifty closes below the strike of 2250, the buyer would exercise his option on the writer who would suffer a loss to the extent of the difference between the strike price and Nifty close. The loss that can be incurred by the writer of the option is a maximum extent of the strike price (Since the worst that can happen is that the asset price can fall to zero) whereas the maximum profit is limited to the extent of the up-front option premium of Rs.61.70 charged by him.



SYNTHETIC LONG CALL: BUY STOCK, BUY PUT

Mr. XYZ is bullish about ABC Ltd stock. He buys ABC Ltd. at current market price of Rs. 4000 on 4th July. To protect against fall in the price of ABC Ltd. (his risk), he buys an ABC Ltd. Put option with a strike price Rs. 3900 (OTM) at a premium of Rs. 143.80 expiring on 31st July.

Strategy : Buy Stock + Buy Put Option		
Buy Stock (Mr. XYZ pays)	Current Market Price of ABC Ltd. (Rs.)	4000
	Strike Price (Rs.)	3900
Buy Put (Mr. XYZ pays)	Premium (Rs.)	143.80
	Break Even Point (Rs.) (Put Strike Price + Put Premium + Stock Price – Put Strike Price)*	4143.80

When to use: When ownership is desired of stock yet investor is concerned about near-term downside risk. The outlook is **conservatively bullish**.

Risk: Losses limited to Stock price + Put Premium – Put Strike price

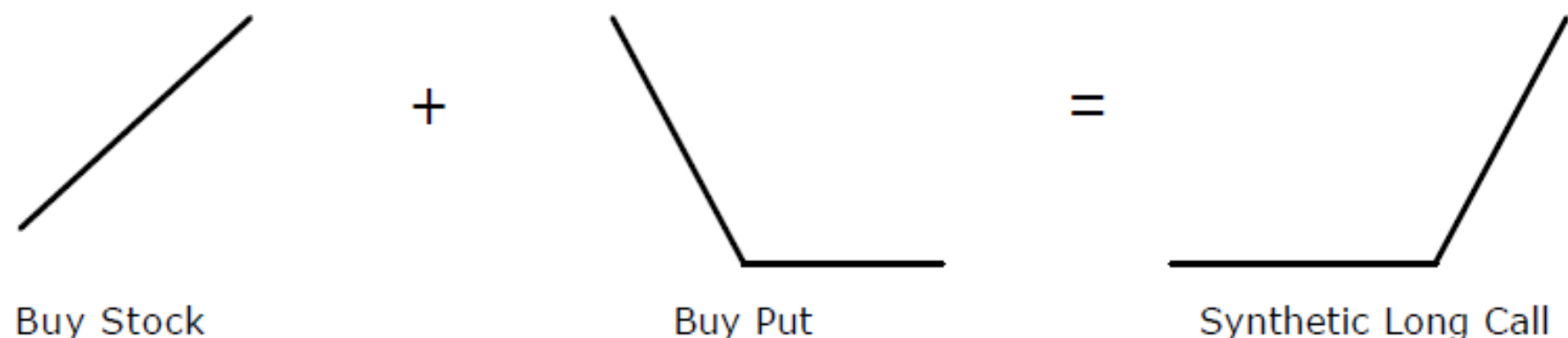
Reward: Profit potential is unlimited.

Break-even Point: Put Strike Price + Put Premium + Stock Price – Put Strike Price

The payoff schedule

ABC Ltd. closes at (Rs.) on expiry	Payoff from the Stock (Rs.)	Net Payoff from the Put Option (Rs.)	Net Payoff (Rs.)
3400.00	-600.00	356.20	-243.80
3600.00	-400.00	156.20	-243.80
3800.00	-200.00	-43.80	-243.80
4000.00	0	-143.80	-143.80
4143.80	143.80	-143.80	0
4200.00	200.00	-143.80	56.20
4400.00	400.00	-143.80	256.20
4600.00	600.00	-143.80	456.20
4800.00	800.00	-143.80	656.20

The payoff chart (Synthetic Long Call)



ANALYSIS:

- This is a low risk strategy. This is a strategy which limits the loss in case of fall in market, but the potential profit remains unlimited when the stock price rises.
- A good strategy when you buy a stock for medium or long term, with the aim of protecting any downside risk. The pay-off resembles a Call Option buy and is therefore called as Synthetic Long Call.

Note: The strategy is not Buy Call Option. Here you have taken an exposure to an underlying stock with the aim of holding it and reaping the benefits of price rise, dividends, bonus rights etc. and at the same time insuring against an adverse price movement.

Covered Call

- You own shares in a company which you feel may rise but not much in the near term and you would still like to earn an income from the shares.
- The covered call is a strategy in which an investor Sells a Call option on a stock he owns (netting him a premium). The Call Option which is sold is usually an OTM Call. The Call would not get exercised unless the stock price increases above the strike price. Till then the investor in the stock (Call seller) can retain the Premium with him. This becomes his income from the stock.
- This strategy is usually adopted by a stock owner who is **Neutral to moderately Bullish** about the stock.
- This strategy is called as a Covered Call strategy because the Call sold is backed by a stock owned by the Call Seller (investor).

COVERED CALL

Strategy : Buy Stock + Sell Call Option		
Mr. A buys the stock XYZ Ltd.	Market Price (Rs.)	3850
Call Options	Strike Price (Rs.)	4000
Mr. A receives	Premium (Rs.)	80
	Break Even Point (Rs.) (Stock Price paid - Premium Received)	3770

Risk: If the Stock Price falls to zero, the investor loses the entire value of the Stock but retains the premium, since the Call will not be exercised against him. So maximum risk = Stock Price Paid - Call Premium

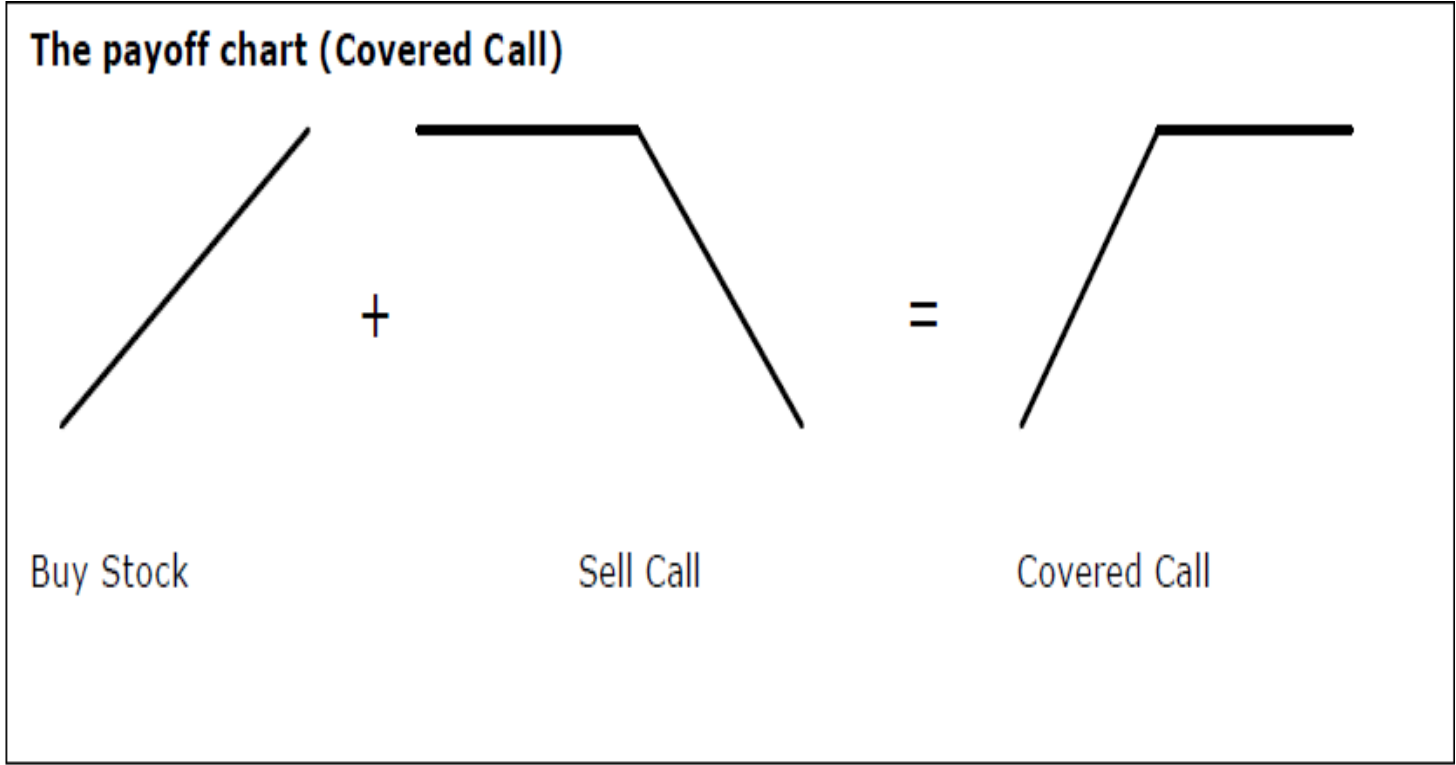
Upside capped at the Strike price plus the Premium received. So if the Stock rises beyond the Strike price the investor (Call seller) gives up all the gains on the stock.

Reward: Limited to (Call Strike Price - Stock Price paid) + Premium received

Breakeven: Stock Price paid - Premium Received

The Payoff Schedule

XYZ Ltd. price closes at (Rs.)	Net Payoff (Rs.)
3600	-170
3700	-70
3740	-30
3770	0
3800	30
3900	130
4000	230
4100	230
4200	230
4300	230



Long Combo: Sell a put and buy a call (OTM)

- A Long Combo is a **Bullish** strategy. If an investor is expecting the price of a stock to move up, he can do a Long Combo strategy.
- It involves selling an OTM (lower strike) Put and buying an OTM (higher strike) Call. This strategy simulates the action of buying a stock (or a futures) but at a fraction of the stock price.
- It is an inexpensive trade, similar in pay-off to Long Stock, except there is a gap between the strikes.

LONG COMBO : SELL A PUT, BUY A CALL

A stock ABC Ltd. is trading at Rs. 450. Mr. XYZ is bullish on the stock. But does not want to invest Rs. 450. He does a Long Combo. He sells a Put option with a strike price Rs. 400 at a premium of Rs. 1.00 and buys a Call Option with a strike price of Rs. 500 at a premium of Rs. 2. The net cost of the strategy (net debit) is Rs. 1.

Strategy : Sell a Put + Buy a Call		
ABC Ltd.	Current Market Price (Rs.)	450
Sells Put	Strike Price (Rs.)	400
Mr. XYZ receives	Premium (Rs.)	1.00
Buys Call	Strike Price (Rs.)	500
Mr. XYZ pays	Premium (Rs.)	2.00
	Net Debit (Rs.)	1.00
	Break Even Point (Rs.) (Higher Strike + Net Debit)	Rs. 501

The payoff schedule

ABC Ltd. closes at (Rs.)	Net Payoff from the Put Sold (Rs.)	Net Payoff from the Call purchased (Rs.)	Net Payoff (Rs.)
700	1	198	199
650	1	148	149
600	1	98	99
550	1	48	49
501	1	-1	0
500	1	-2	-1
450	1	-2	-1
400	1	-2	-1
350	-49	-2	-51
300	-99	-2	-101
250	-149	-2	-151

The payoff chart (Long Combo)



For a small investment of Re. 1 (net debit), the returns can be very high in a Long Combo, but only if the stock moves up. Otherwise the potential losses can also be high.

Protective Call/ Synthetic Long Put

- This is a strategy wherein an investor has gone short on a stock and buys a call to hedge. This is an opposite of Synthetic Call.
- An investor shorts a stock and buys an ATM or slightly OTM Call. The net effect of this is that the investor creates a pay-off like a Long Put, but instead of having a net debit (paying premium) for a Long Put, he creates a net credit (receives money on shorting the stock).
- In case the stock price falls the investor gains in the downward fall in the price. However, incase there is an unexpected rise in the price of the stock the loss is limited.
- The pay-off from the Long Call will increase thereby compensating for the loss in value of the short stock position. This strategy hedges the upside in the stock position while retaining downside profit potential.

PROTECTIVE CALL /SYNTHETIC LONG PUT

Suppose ABC Ltd. is trading at Rs. 4457 in June. An investor Mr. A buys a Rs 4500 call for Rs. 100 while shorting the stock at Rs. 4457. The net credit to the investor is Rs. 4357 (Rs. 4457 – Rs. 100).

Strategy : Short Stock + Buy Call Option		
Sells Stock (Mr. A receives)	Current Market Price (Rs.)	4457
Buys Call	Strike Price (Rs.)	4500
Mr. A pays	Premium (Rs.)	100
	Break Even Point (Rs.) (Stock Price – Call Premium)	4357

When to Use: If the investor is of the view that the markets will go down **(bearish)** but wants to protect against any unexpected rise in the price of the stock.

Risk: Limited. Maximum Risk is Call Strike Price – Stock Price + Premium

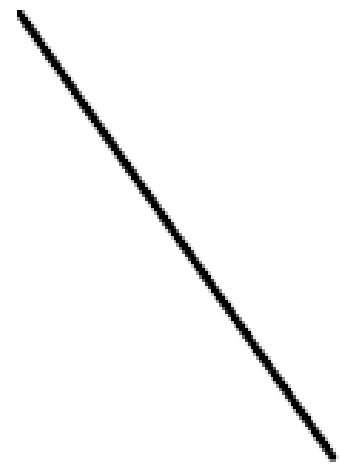
Reward: Maximum is Stock Price – Call Premium

Breakeven: Stock Price – Call Premium

The payoff schedule

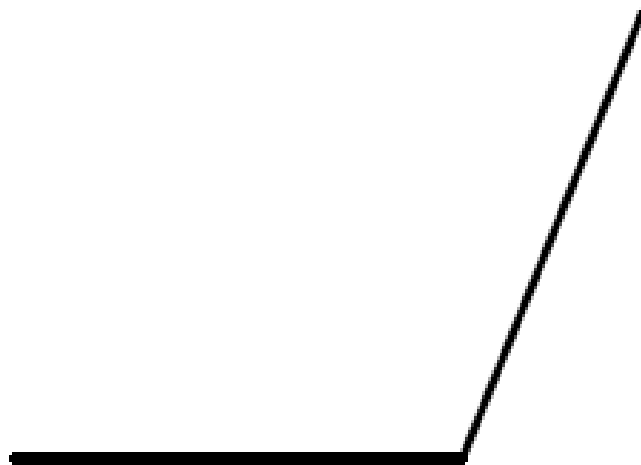
ABC Ltd. closes at (Rs.)	Payoff from the stock (Rs.)	Net Payoff from the Call Option (Rs.)	Net Payoff (Rs.)
4100	357	-100	257
4150	307	-100	207
4200	257	-100	157
4300	157	-100	57
4350	107	-100	7
4357	100	-100	0
4400	57	-100	-43
4457	0	-100	-100
4600	-143	0	-143
4700	-243	100	-143
4800	-343	200	-143
4900	-443	300	-143
5000	-543	400	-143

The payoff chart (Synthetic Long Put)



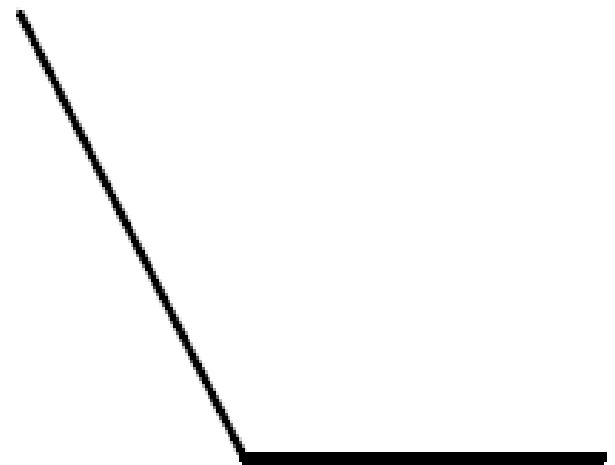
Sell Stock

+



Buy Call

=



Synthetic Long Put

Covered Put

- This strategy is opposite to a Covered Call. A Covered Call is a neutral to bullish strategy, whereas a Covered Put is a neutral to **Bearish** strategy.
- One should go for this strategy when someone feels the price of a stock is going to remain range bound or move down.
- Covered Put writing involves a short in a stock along with a short Put on the options on the stock.
- If the stock price does not change, the investor gets to keep the Premium. He can use this strategy as an income in a neutral market.

COVERED PUT

Suppose ABC Ltd. is trading at Rs 4500 in June. An investor, Mr. A, shorts Rs 4300 Put by selling a July Put for Rs. 24 while shorting an ABC Ltd. stock. The net credit received by Mr. A is Rs. 4500 + Rs. 24 = Rs. 4524.

Strategy : Short Stock + Short Put Option		
Sells Stock (Mr. A receives)	Current Market Price (Rs.)	4500
Sells Put	Strike Price (Rs.)	4300
Mr. A receives	Premium (Rs.)	24
	Break Even Point (Rs.) (Sale price of Stock + Put Premium)	4524

When to Use: If the investor is of the view that the markets are **moderately bearish**.

Risk: Unlimited if the price of the stock rises substantially

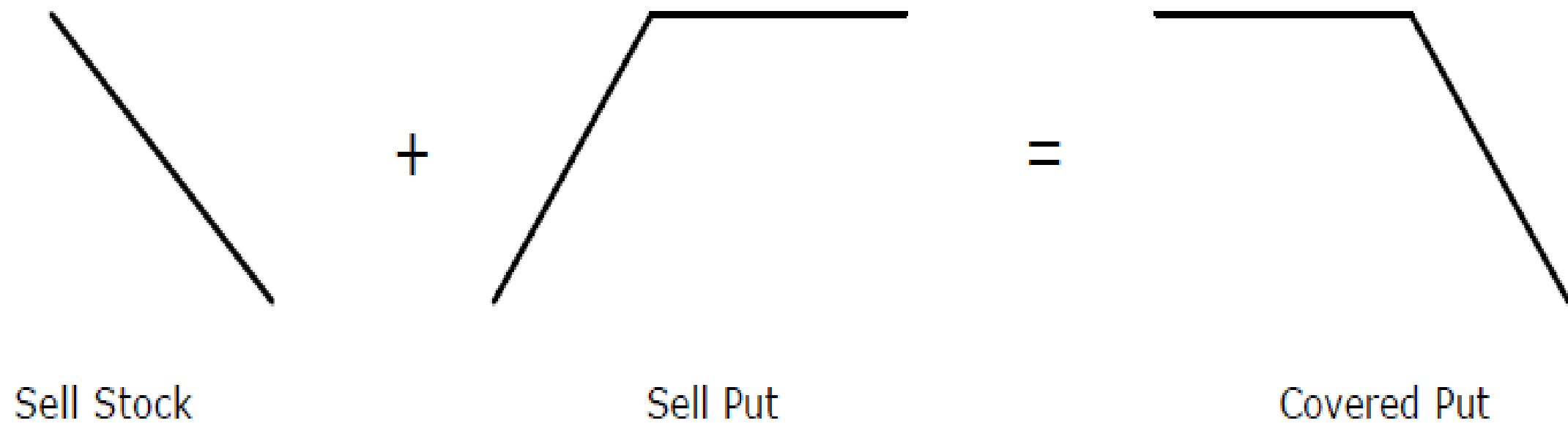
Reward: Maximum is (Sale Price of the Stock – Strike Price) + Put Premium

Breakeven: Sale Price of Stock + Put Premium

The payoff schedule

ABC Ltd. closes at (Rs.)	Payoff from the stock (Rs.)	Net Payoff from the Put Option (Rs.)	Net Payoff (Rs.)
4000	500	-276	224
4100	400	-176	224
4200	300	-76	224
4300	200	24	224
4400	100	24	124
4450	50	24	74
4500	0	24	24
4524	-24	24	0
4550	-50	24	-26
4600	-100	24	-76
4635	-135	24	-111
4650	-160	24	-136

The payoff chart (Covered Put)



LONG STRADDLE

- A Straddle is a volatility strategy and is used when the stock price / index is expected to show large movements.
- This strategy involves buying a call as well as put on the same stock / index for the same maturity and strike price, to take advantage of a movement in either direction, a soaring or plummeting value of the stock / index.
- If the price of the stock/ index increases, the call is exercised while the put expires worthless and if the price of the stock / index decreases, the put is exercised, the call expires worthless. Either way if the stock / index shows volatility to cover the cost of the trade, profits are to be made.

LONG STRADDLE

When to Use: The investor thinks that the underlying stock / index will experience **significant volatility** in the near term.

Risk: Limited to the initial premium paid.

Reward: Unlimited

Breakeven:

- Upper Breakeven Point = Strike Price of Long Call + Net Premium Paid
- Lower Breakeven Point = Strike Price of Long Put - Net Premium Paid

Example

Suppose Nifty is at 4450 on 27th April. An investor, Mr. A enters a long straddle by buying a May Rs 4500 Nifty Put for Rs. 85 and a May Rs. 4500 Nifty Call for Rs. 122. The net debit taken to enter the trade is Rs 207, which is also his maximum possible loss.

Strategy : Buy Put + Buy Call

Nifty index	Current Value	4450
Call and Put	Strike Price (Rs.)	4500
Mr. A pays	Total Premium (Call + Put) (Rs.)	207
	Break Even Point (Rs.)	4707(U)
	(Rs.)	4293(L)

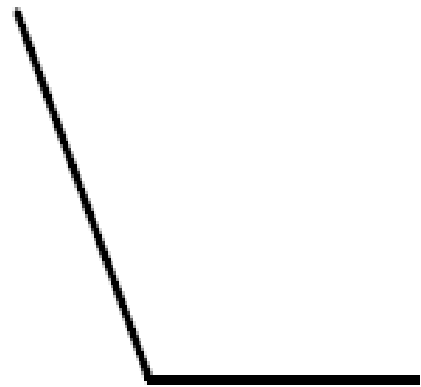
The call option premium is 122

The put option premium is 85

The Payoff Schedule

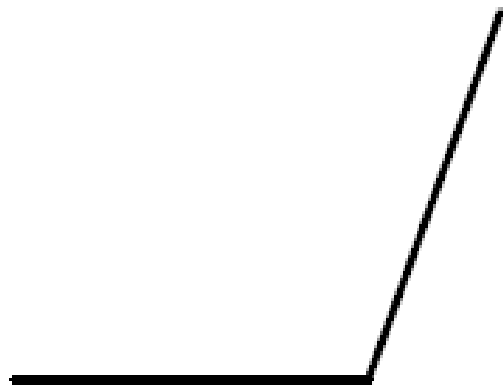
On expiry Nifty closes at	Net Payoff from Put purchased (Rs.)	Net Payoff from Call purchased (Rs.)	Net Payoff (Rs.)
3800	615	-122	493
3900	515	-122	393
4000	415	-122	293
4100	315	-122	193
4200	215	-122	93
4234	181	-122	59
4293	122	-122	0
4300	115	-122	-7
4400	15	-122	-107
4500	-85	-122	-207
4600	-85	-22	-107
4700	-85	78	-7
4707	-85	85	0
4766	-85	144	59
4800	-85	178	93
4900	-85	278	193
5000	-85	378	293
5100	-85	478	393
5200	-85	578	493
5300	-85	678	593

The payoff chart (Long Straddle)



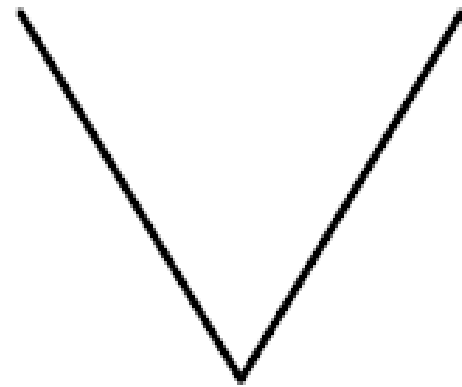
Buy Put

+



Buy Call

=



Long Straddle

SHORT STRADDLE

When to Use: The investor thinks that the underlying stock / index will experience **very little volatility** in the near term.

Risk: Unlimited

Reward: Limited to the premium received

Breakeven:

- Upper Breakeven Point = Strike Price of Short Call + Net Premium Received
- Lower Breakeven Point = Strike Price of Short Put - Net Premium Received

Example

Suppose Nifty is at 4450 on 27th April. An investor, Mr. A, enters into a short straddle by selling a May Rs 4500 Nifty Put for Rs. 85 and a May Rs. 4500 Nifty Call for Rs. 122. The net credit received is Rs. 207, which is also his maximum possible profit.

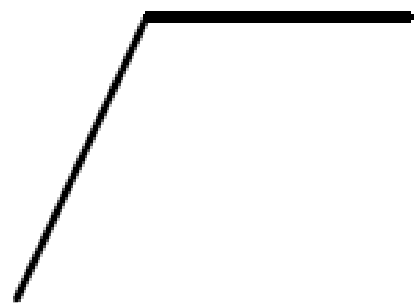
Strategy : Sell Put + Sell Call		
Nifty index	Current Value	4450
Call and Put	Strike Price (Rs.)	4500
Mr. A receives	Total Premium (Call + Put) (Rs.)	207
	Break Even Point (Rs.)*	4707(U)
	(Rs.)*	4293(L)

* From buyer's point of view

The payoff schedule

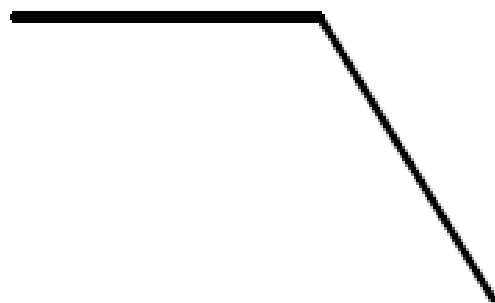
On expiry Nifty closes at	Net Payoff from Put Sold (Rs.)	Net Payoff from Call Sold (Rs.)	Net Payoff (Rs.)
3800	-615	122	-493
3900	-515	122	-393
4000	-415	122	-293
4100	-315	122	-193
4200	-215	122	-93
4234	-181	122	-59
4293	-122	122	0
4300	-115	122	7
4400	-15	122	107
4500	85	122	207
4600	85	22	107
4700	85	-78	7
4707	85	-85	0
4766	85	-144	-59
4800	85	-178	-93
4900	85	-278	-193
5000	85	-378	-293

The payoff chart (Short Straddle)



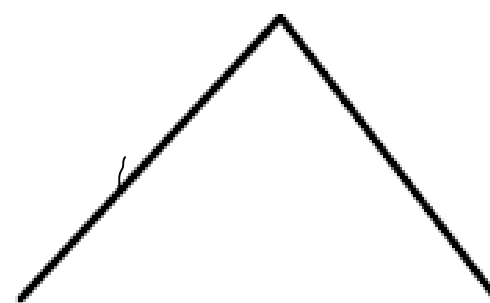
Sell Put

+



Sell Call

=



Short Straddle

LONG STRANGLE

- A Strangle is a slight modification to the Straddle to make it cheaper to execute.
- This strategy involves the simultaneous buying of a slightly out-of-the-money (OTM) put and a slightly out-of-the-money (OTM) call of the same underlying stock / index and expiration date.
- Since OTM options are purchased for both Calls and Puts it makes the cost of executing a Strangle cheaper as compared to a Straddle. Since the initial cost of a Strangle is cheaper than a Straddle, the returns could potentially be higher.

LONG STRANGLE

When to Use: The investor thinks that the underlying stock / index will experience **very high levels of volatility** in the near term.

Risk: Limited to the initial premium paid

Reward: Unlimited

Breakeven:

- Upper Breakeven Point = Strike Price of Long Call + Net Premium Paid
- Lower Breakeven Point = Strike Price of Long Put - Net Premium Paid

Example

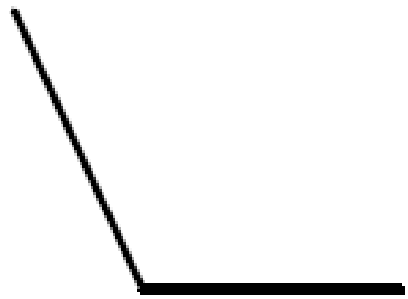
Suppose Nifty is at 4500 in May. An investor, Mr. A, executes a Long Strangle by buying a Rs. 4300 Nifty Put for a premium of Rs. 23 and a Rs 4700 Nifty Call for Rs 43. The net debit taken to enter the trade is Rs. 66, which is also his maximum possible loss.

Strategy : Buy OTM Put + Buy OTM Call		
Nifty index	Current Value	4500
Buy Call Option	Strike Price (Rs.)	4700
Mr. A pays	Premium (Rs.)	43
	Break Even Point (Rs.)	4766
Buy Put Option	Strike Price (Rs.)	4300
Mr. A pays	Premium (Rs.)	23
	Break Even Point (Rs.)	4234

The payoff schedule

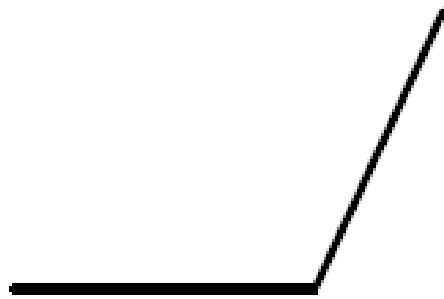
On expiry Nifty closes at	Net Payoff from Put purchased (Rs.)	Net Payoff from Call purchased (Rs.)	Net Payoff (Rs.)
3800	477	-43	434
3900	377	-43	334
4000	277	-43	234
4100	177	-43	134
4200	77	-43	34
4234	43	-43	0
4300	-23	-43	-66
4400	-23	-43	-66
4500	-23	-43	-66
4600	-23	-43	-66
4700	-23	-43	-66
4766	-23	23	0
4800	-23	57	34
4900	-23	157	134
5000	-23	257	234
5100	-23	357	334
5200	-23	457	434
5300	-23	557	534

The payoff chart (Long Strangle)



Buy OTM Put

+



Buy OTM Call

=



Long Strangle

SHORT STRANGLE

- A Short Strangle is a slight modification to the Short Straddle. It tries to improve the profitability of the trade for the Seller of the options by widening the breakeven points so that there is a much greater movement required in the underlying stock / index, for the Call and Put option to be worth exercising.
- This strategy involves the simultaneous selling of a slightly out-of-the-money (OTM) put and a slightly out-of-the-money (OTM) call of the same underlying stock and expiration date. This typically means that since OTM call and put are sold, the net credit received by the seller is less as compared to a Short Straddle, but the breakeven points are also widened.

SHORT STRANGLE

When to Use: This options trading strategy is taken when the options investor thinks that the underlying stock will experience **little volatility** in the near term.

Risk: Unlimited

Reward: Limited to the premium received

Breakeven:

- Upper Breakeven Point = Strike Price of Short Call + Net Premium Received
- Lower Breakeven Point = Strike Price of Short Put - Net Premium Received

Example

Suppose Nifty is at 4500 in May. An investor, Mr. A, executes a Short Strangle by selling a Rs. 4300 Nifty Put for a premium of Rs. 23 and a Rs. 4700 Nifty Call for Rs 43. The net credit is Rs. 66, which is also his maximum possible gain.

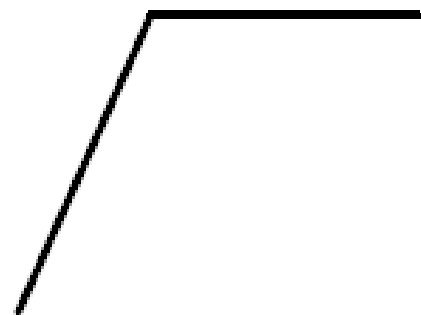
Strategy : Sell OTM Put + Sell OTM Call

Nifty index	Current Value	4500
Sell Call Option	Strike Price (Rs.)	4700
Mr. A receives	Premium (Rs.)	43
	Break Even Point (Rs.)	4766
Sell Put Option	Strike Price (Rs.)	4300
Mr. A receives	Premium (Rs.)	23
	Break Even Point (Rs.)	4234

The payoff schedule

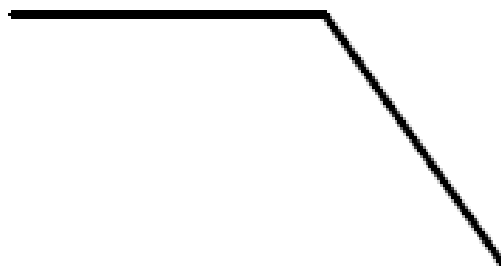
On expiry Nifty closes at	Net Payoff from Put sold (Rs.)	Net Payoff from Call sold (Rs.)	Net Payoff (Rs.)
3800	-477	43	-434
3900	-377	43	-334
4000	-277	43	-234
4100	-177	43	-134
4200	-77	43	-34
4234	-43	43	0
4300	23	43	66
4400	23	43	66
4500	23	43	66
4600	23	43	66
4700	23	43	66
4766	23	-23	0
4800	23	-57	-34
4900	23	-157	-134
5000	23	-257	-234
5100	23	-357	-334
5200	23	-457	-434
5300	23	-557	-534

The payoff chart (Short Strangle)



Sell OTM Put

+



Sell OTM Call

=



Short Strangle

COLLAR

- A Collar is similar to Covered Call but involves another leg – buying a Put to insure against the fall in the price of the stock. It is a Covered Call with a limited risk.
- So, a Collar is buying a stock, insuring against the downside by buying a Put and then financing (partly) the Put by selling a Call.
- The put generally is ATM and the call is OTM having the same expiration month and must be equal in number of shares. This is a low-risk strategy since the Put prevents downside risk. However, do not expect unlimited rewards since the Call prevents that.
- It is a strategy to be adopted when the investor is **conservatively bullish**.

COLLAR

When to Use: The collar is a good strategy to use if the investor is writing covered calls to earn premiums but wishes to protect himself from an unexpected sharp drop in the price of the underlying security.

Risk: Limited

Reward: Limited

Breakeven: $\text{Purchase Price of Underlying} - \text{Call Premium} + \text{Put Premium}$

Example

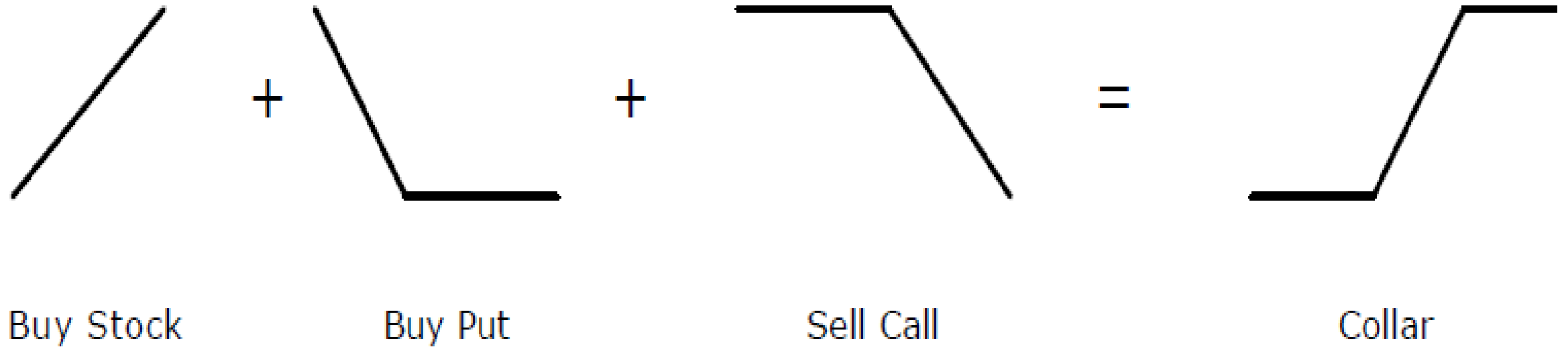
Suppose an investor Mr. A buys or is holding ABC Ltd. currently trading at Rs. 4758. He decides to establish a collar by writing a Call of strike price Rs. 5000 for Rs. 39 while simultaneously purchasing a Rs. 4700 strike price Put for Rs. 27.

Since he pays Rs. 4758 for the stock ABC Ltd., another Rs. 27 for the Put but receives Rs. 39 for selling the Call option, his total investment is Rs. 4746.

Strategy : Buy Stock + Buy Put + Sell Call		
ABC Ltd.	Current Market Price (Rs.)	4758
Sell Call Option	Strike Price (Rs.)	5000
Mr. A Receives	Premium (Rs.)	39
Buy Put Option	Strike Price (Rs.)	4700
Mr. A Pays	Premium (Rs.)	27
	Net Premium Received(Rs.)	12
	Break Even Point (Rs.)	4746

ABC Ltd. closes at (Rs.)	Payoff from Call Sold (Rs.)	Payoff from Put Purchased (Rs.)	Payoff from stock ABC Ltd.	Net payoff (Rs.)
4400	39	273	-358	-46
4450	39	223	-308	-46
4500	39	173	-258	-46
4600	39	73	-158	-46
4700	39	-27	-58	-46
4750	39	-27	-8	4
4800	39	-27	42	54
4850	39	-27	92	104
4858	39	-27	100	112
4900	39	-27	142	154
4948	39	-27	190	202
5000	39	-27	242	254
5050	-11	-27	292	254
5100	-61	-27	342	254
5150	-111	-27	392	254

The payoff chart (Collar)



BULL CALL SPREAD STRATEGY

- A bull call spread is constructed by buying an in-the-money (ITM) call option and selling another out-of-the-money (OTM) call option.
- Often the call with the lower strike price will be in-the-money while the Call with the higher strike price is out-of-the-money. Both calls must have the same underlying security and expiration month.
- The net effect of the strategy is to bring down the cost and breakeven on a Buy Call (Long Call) Strategy.
- This strategy is exercised when investor is moderately bullish to bullish, because the investor will make a profit only when the stock price / index rises.

BULL CALL SPREAD STRATEGY

When to Use: Investor is moderately bullish.

Risk: Limited to any initial premium paid in establishing the position. Maximum loss occurs where the underlying falls to the level of the lower strike or below.

Reward: Limited to the difference between the two strikes minus net premium cost. Maximum profit occurs where the underlying rises to the level of the higher strike or above

Break-Even-Point (BEP):
Strike Price of Purchased call
+ Net Debit Paid

Example:

Mr. XYZ buys a Nifty Call with a Strike price Rs. 4100 at a premium of Rs. 170.45 and he sells a Nifty Call option with a strike price Rs. 4400 at a premium of Rs. 35.40. The net debit here is Rs. 135.05 which is also his maximum loss.

Strategy : Buy a Call with a lower strike (ITM) + Sell a Call with a higher strike (OTM)		
Nifty index	Current Value	4191.10
Buy ITM Call Option	Strike Price (Rs.)	4100
Mr. XYZ Pays	Premium (Rs.)	170.45
Sell OTM Call Option	Strike Price (Rs.)	4400
Mr. XYZ Receives	Premium (Rs.)	35.40
	Net Premium Paid (Rs.)	135.05
	Break Even Point (Rs.)	4235.05

The payoff schedule :

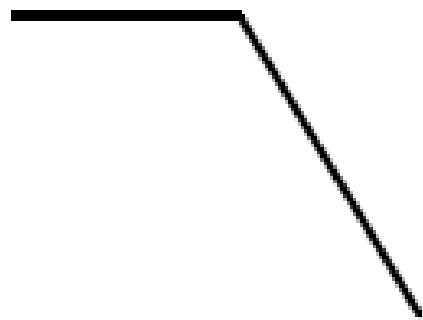
On expiry Nifty Closes at	Net Payoff from Call Buy (Rs.)	Net Payoff from Call Sold (Rs.)	Net Payoff (Rs.)
3500.00	-170.45	35.40	-135.05
3600.00	-170.45	35.40	-135.05
3700.00	-170.45	35.40	-135.05
3800.00	-170.45	35.40	-135.05
3900.00	-170.45	35.40	-135.05
4000.00	-170.45	35.40	-135.05
4100.00	-170.45	35.40	-135.05
4200.00	-70.45	35.40	-35.05
4235.05	-35.40	35.40	0
4300.00	29.55	35.40	64.95
4400.00	129.55	35.40	164.95
4500.00	229.55	-64.60	164.95
4600.00	329.55	-164.60	164.95
4700.00	429.55	-264.60	164.95
4800.00	529.55	-364.60	164.95
4900.00	629.55	-464.60	164.95
5000.00	729.55	-564.60	164.95
5100.00	829.55	-664.60	164.95
5200.00	929.55	-764.60	164.95

The payoff chart (Bull Call Spread)



Buy lower strike Call

+



Sell OTM Call

=



Bull Call Spread

BULL PUT SPREAD STRATEGY

When to Use: When the investor is **moderately bullish**.

Risk: Limited. Maximum loss occurs where the underlying falls to the level of the lower strike or below

Reward: Limited to the net premium credit. Maximum profit occurs where underlying rises to the level of the higher strike or above.

Breakeven: Strike Price of Short Put - Net Premium Received

Example:

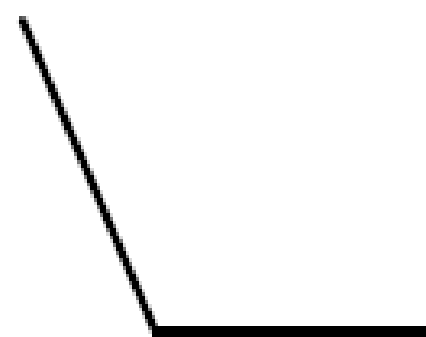
Mr. XYZ sells a Nifty Put option with a strike price of Rs. 4000 at a premium of Rs. 21.45 and buys a further OTM Nifty Put option with a strike price Rs. 3800 at a premium of Rs. 3.00 when the current Nifty is at 4191.10, with both options expiring on 31st July.

Strategy : Sell a Put + Buy a Put		
Nifty Index	Current Value	4191.10
Sell Put Option	Strike Price (Rs.)	4000
Mr. XYZ Receives	Premium (Rs.)	21.45
Buy Put Option	Strike Price (Rs.)	3800
Mr. XYZ Pays	Premium (Rs.)	3.00
	Net Premium Received (Rs.)	18.45
	Break Even Point (Rs.)	3981.55

The payoff schedule

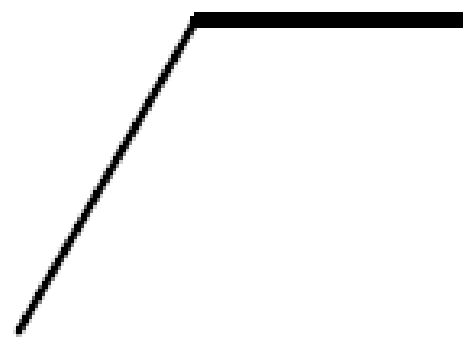
On expiry Nifty Closes at	Net Payoff from Put Buy (Rs.)	Net Payoff from Put Sold (Rs.)	Net Payoff (Rs.)
3500.00	297.00	-478.55	-181.55
3600.00	197.00	-378.55	-181.55
3700.00	97.00	-278.55	-181.55
3800.00	-3.00	-178.55	-181.55
3900.00	-3.00	-78.55	-81.55
3981.55	-3.00	3.00	0.00
4000.00	-3.00	21.45	18.45
4100.00	-3.00	21.45	18.45
4200.00	-3.00	21.45	18.45
4300.00	-3.00	21.45	18.45
4400.00	-3.00	21.45	18.45
4500.00	-3.00	21.45	18.45
4600.00	-3.00	21.45	18.45
4700.00	-3.00	21.45	18.45
4800.00	-3.00	21.45	18.45

The payoff chart (Bull Put Spread)



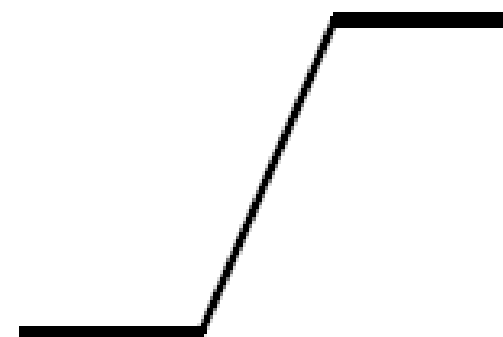
Buy lower strike Put

+



Sell OTM Put

=



Bull Put Spread

Analysis

- The Bull Call Spread Strategy has brought the breakeven point down (if only the Rs. 4100 strike price Call was purchased the breakeven point would have been Rs. 4270.45),
- Reduced the cost of the trade (if only the Rs. 4100 strike price Call was purchased the cost of the trade would have been Rs. 170.45),
- reduced the loss on the trade (if only the Rs. 4150 strike price Call was purchased the loss would have been Rs. 170.45 i.e. the premium of the Call purchased).
- However, the strategy also has limited gains and is therefore ideal when markets are moderately bullish.

BEAR CALL SPREAD STRATEGY

- The Bear Call Spread strategy can be adopted when the investor feels that the stock / index is either range bound or falling. The concept is to protect the downside of a Call Sold by buying a Call of a higher strike price to insure the Call sold.
- In this strategy the investor receives a net credit because the Call he buys is of a higher strike price than the Call sold.
- The strategy requires the investor to buy out-of-the-money (OTM) call options while simultaneously selling in-the-money (ITM) call options on the same underlying stock index.
- This strategy can also be done with both OTM calls with the Call purchased being higher OTM strike than the Call sold.

BEAR CALL SPREAD STRATEGY

When to use: When the investor is mildly bearish on market.

Risk: Limited to the difference between the two strikes minus the net premium.

Reward: Limited to the net premium received for the position i.e., premium received for the short call minus the premium paid for the long call.

Break Even Point:
Lower Strike + Net credit

Example:

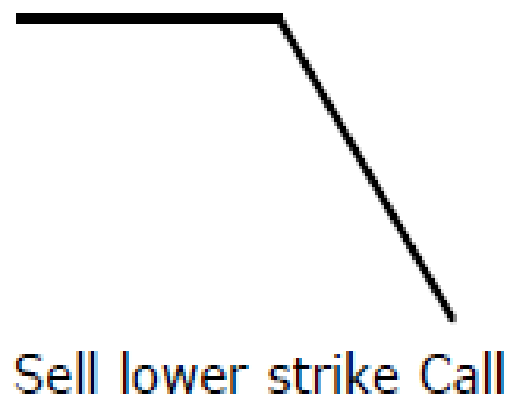
Mr. XYZ is bearish on Nifty. He sells an ITM call option with strike price of Rs. 2600 at a premium of Rs. 154 and buys an OTM call option with strike price Rs. 2800 at a premium of Rs. 49.

**Strategy : Sell a Call with a lower strike (ITM)
+ Buy a Call with a higher strike (OTM)**

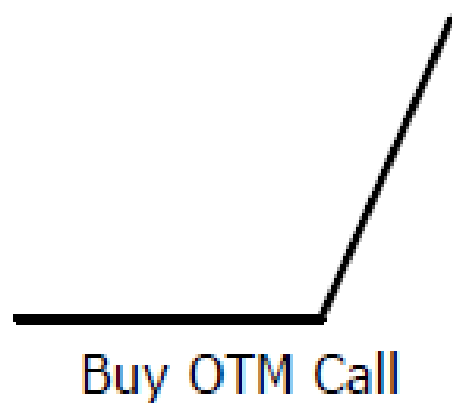
Nifty index	Current Value	2694
Sell ITM Call Option	Strike Price (Rs.)	2600
Mr. XYZ receives	Premium (Rs.)	154
Buy OTM Call Option	Strike Price (Rs.)	2800
Mr. XYZ pays	Premium (Rs.)	49
	Net premium received (Rs.)	105
	Break Even Point (Rs.)	2705

On expiry Nifty Closes at	Net Payoff from Call Sold (Rs.)	Net Payoff from Call bought (Rs.)	Net Payoff (Rs.)
2100	154	-49	105
2200	154	-49	105
2300	154	-49	105
2400	154	-49	105
2500	154	-49	105
2600	154	-49	105
2700	54	-49	5
2705	49	-49	0
2800	-46	-49	-95
2900	-146	51	-95
3000	-246	151	-95
3100	-346	251	-95
3200	-446	351	-95
3300	-546	451	-95

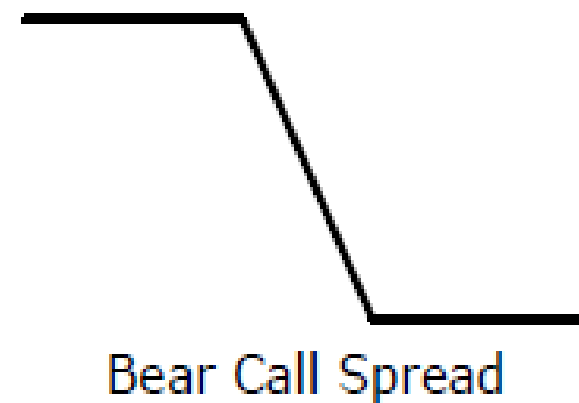
The payoff chart (Bear Call Spread)



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BEAR PUT SPREAD STRATEGY

When to use: When you are **moderately bearish** on market direction

Risk: Limited to the net amount paid for the spread. i.e. the premium paid for long position less premium received for short position.

Reward: Limited to the difference between the two strike prices minus the net premium paid for the position.

Break Even Point: Strike Price of Long Put - Net Premium Paid

Example:

Nifty is presently at 2694. Mr. XYZ expects Nifty to fall. He buys one Nifty ITM Put with a strike price Rs. 2800 at a premium of Rs. 132 and sells one Nifty OTM Put with strike price Rs. 2600 at a premium Rs. 52.

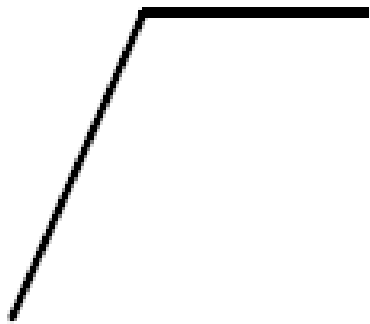
Strategy : BUY A PUT with a higher strike (ITM) + SELL A PUT with a lower strike (OTM)

Nifty index	Current Value	2694
Buy ITM Put Option	Strike Price (Rs.)	2800
Mr. XYZ pays	Premium (Rs.)	132
Sell OTM Put Option	Strike Price (Rs.)	2600
Mr. XYZ receives	Premium (Rs.)	52
	Net Premium Paid (Rs.)	80
	Break Even Point (Rs.)	2720

The payoff schedule

On expiry Nifty closes at	Net Payoff from Put Buy (Rs.)	Net Payoff from Put Sold (Rs.)	Net payoff (Rs.)
2200	468	-348	120
2300	368	-248	120
2400	268	-148	120
2500	168	-48	120
2600	68	52	120
2720	-52	52	0
2700	-32	52	20
2800	-132	52	-80
2900	-132	52	-80
3000	-132	52	-80
3100	-132	52	-80

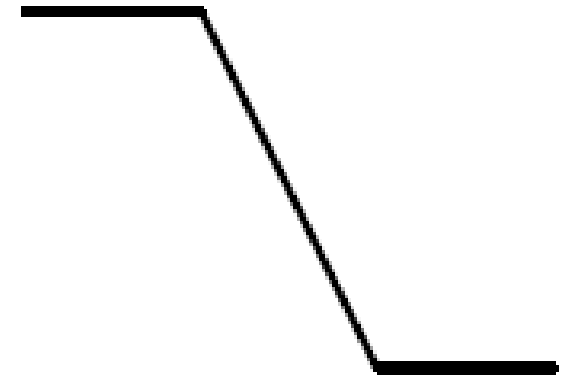
The payoff chart (Bear Put Spread)



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Sell lower strike Put

Buy Put

Bear Put Spread

LONG CALL BUTTERFLY

When to use: When the investor is neutral on market direction and bearish on volatility.

Risk Net debit paid.

Reward Difference between adjacent strikes minus net debit

Break Even Point:

Upper Breakeven Point =
Strike Price of Higher
Strike Long Call - Net
Premium Paid

Lower Breakeven Point =
Strike Price of Lower
Strike Long Call + Net
Premium Paid

Example:

Nifty is at 3200. Mr. XYZ expects very little movement in Nifty. He sells 2 ATM Nifty Call Options with a strike price of Rs. 3200 at a premium of Rs. 97.90 each, buys 1 ITM Nifty Call Option with a strike price of Rs. 3100 at a premium of Rs. 141.55 and buys 1 OTM Nifty Call Option with a strike price of Rs. 3300 at a premium of Rs. 64. The Net debit is Rs. 9.75.

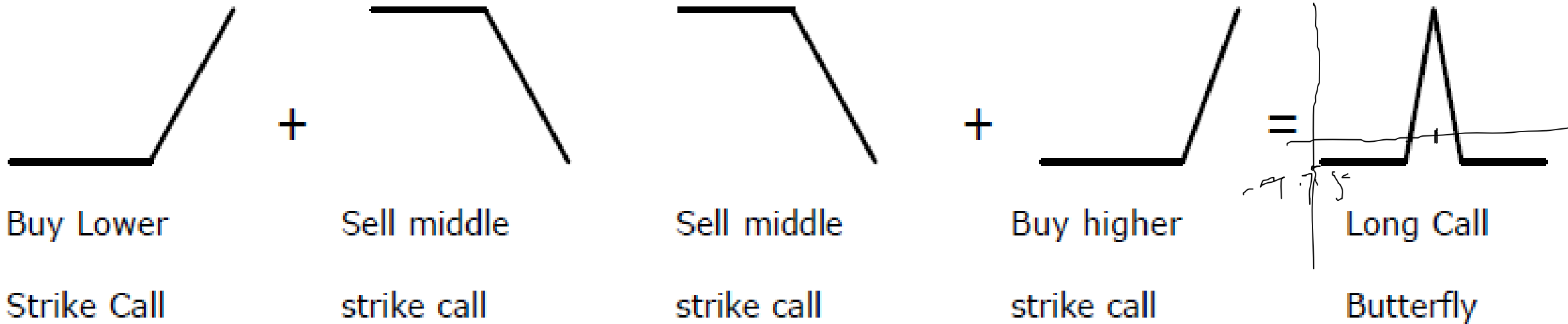
STRATEGY : SELL 2 ATM CALL, BUY 1 ITM CALL OPTION AND BUY 1 OTM CALL OPTION		
Nifty index	Current Value	3200
Sell 2 ATM Call Option	Strike Price (Rs.)	3200
Mr. XYZ receives	Premium (Rs.)	195.80
Buy 1 ITM Call Option	Strike Price (Rs.)	3100
Mr. XYZ pays	Premium (Rs.)	141.55
Buy 1 OTM Call Option	Strike Price (Rs.)	3300
Mr. XYZ pays	Premium (Rs.)	64
	Break Even Point (Rs.)	3290.25
	Break Even Point (Lower) (Rs.)	3109.75

-9.75

The Payoff Schedule

On expiry Nifty Closes at	Net Payoff from 2 ATM Calls Sold (Rs.)	Net Payoff from 1 ITM Call purchased (Rs.)	Net Payoff from 1 OTM Call purchased (Rs.)	Net Payoff (Rs.)
2700.00	195.80	-141.55	-64	-9.75
2800.00	195.80	-141.55	-64	-9.75
2900.00	195.80	-141.55	-64	-9.75
3000.00	195.80	-141.55	-64	-9.75
3100.00	195.80	-141.55	-64	-9.75
3109.75	195.80	-131.80	-64	0
3200.00	195.80	-41.55	-64	90.25
3290.25	15.30	48.70	-64	0
3300.00	-4.20	58.45	-64	-9.75
3400.00	-204.20	158.45	36	-9.75
3500.00	-404.20	258.45	136	-9.75
3600.00	-604.20	358.45	236	-9.75
3700.00	-804.20	458.45	336	-9.75
3800.00	-1004.20	558.45	436	-9.75
3900.00	-1204.20	658.45	536	-9.75

The payoff chart (Long Call Butterfly)



SHORT CALL BUTTERFLY

Nifty is at 3200. Mr. XYZ expects large volatility in the Nifty irrespective of which direction the movement is, upwards or downwards. Mr. XYZ buys 2 ATM Nifty Call Options with a strike price of Rs. 3200 at a premium of Rs. 97.90 each, sells 1 ITM Nifty Call Option with a strike price of Rs. 3100 at a premium of Rs. 141.55 and sells 1 OTM Nifty Call Option with a strike price of Rs. 3300 at a premium of Rs. 64. The Net Credit is Rs. 9.75.

STRATEGY

BUY 2 ATM CALL OPTIONS, SELL 1 ITM CALL OPTION AND SELL 1 OTM CALL OPTION.

Nifty index	Current Market Price	3200
Buy 2 ATM Call Option	Strike Price (Rs.)	3200
Mr. XYZ pays	Premium (Rs.)	195.80
Sells 1 ITM Call Option	Strike Price (Rs.)	3100
Mr. XYZ receives	Premium (Rs.)	141.55
Sells 1 OTM Call Option	Strike Price (Rs.)	3300
Mr. XYZ receives	Premium (Rs.)	64
	Break Even Point (Upper) (Rs.)	3290.25
	Break Even Point (Lower) (Rs.)	3109.75

When to use: You are neutral on market direction and bullish on volatility. Neutral means that you expect the market to move in either direction - i.e. bullish and bearish.

Risk Limited to the net difference between the adjacent strikes (Rs. 100 in this example) less the premium received for the position.

Reward Limited to the net premium received for the option spread.

Break Even Point:

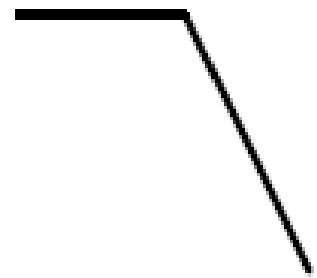
Upper Breakeven Point =
Strike Price of Highest Strike
Short Call - Net Premium
Received

Lower Breakeven Point =
Strike Price of Lowest Strike
Short Call + Net Premium
Received

The Payoff Schedule

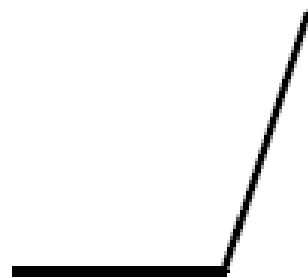
On expiry Nifty Closes at	Net Payoff from 2 ATM Calls Purchased (Rs.)	Net Payoff from 1 ITM Call sold (Rs.)	Net Payoff from 1 OTM Call sold (Rs.)	Net Payoff (Rs.)
2700.00	-195.80	141.55	64.00	9.75
2800.00	-195.80	141.55	64.00	9.75
2900.00	-195.80	141.55	64.00	9.75
3000.00	-195.80	141.55	64.00	9.75
3100.00	-195.80	141.55	64.00	9.75
3109.75	-195.80	131.80	64.00	0
3200.00	-195.80	41.55	64.00	-90.25
3290.25	-15.30	-48.70	64.00	0
3300.00	4.20	-58.45	64.00	9.75
3400.00	204.20	-158.45	-36.00	9.75
3500.00	404.20	-258.45	-136.00	9.75
3600.00	604.20	-358.45	-236.00	9.75
3700.00	804.20	-458.45	-336.00	9.75
3800.00	1004.20	-558.45	-436.00	9.75
3900.00	1204.20	-658.45	-536.00	9.75

The payoff chart (Short Call Butterfly)

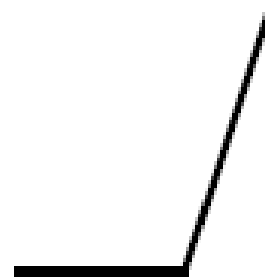


Sell Lower
Strike Call

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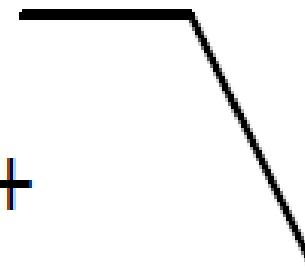


Buy middle
strike call



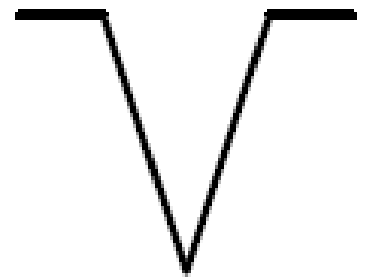
Buy middle
strike call

+



Sell higher
strike call

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Short Call
Butterfly

LONG CALL CONDOR

When to Use: When an investor believes that the underlying market will trade in a range with low volatility until the options expire.

Risk Limited to the minimum of the difference between the lower strike call spread less the higher call spread less the total premium paid for the condor.

Reward Limited. The maximum profit of a long condor will be realized when the stock is trading between the two middle strike prices.

Break Even Point:

Upper Breakeven Point = Highest Strike – Net Debit

Lower Breakeven Point = Lowest Strike + Net Debit

Example: Nifty is at 3600. Mr. XYZ expects little volatility in the Nifty and expects the market to remain rangebound. Mr. XYZ buys 1 ITM Nifty Call Options with a strike price of Rs. 3400 at a premium of Rs. 41.25, sells 1 ITM Nifty Call Option with a strike price of Rs. 3500 at a premium of Rs. 26, sells 1 OTM Nifty Call Option with a strike price of Rs. 3700 at a premium of Rs. 9.80 and buys 1 OTM Nifty Call Option with a strike price of Rs. 3800 at a premium of Rs. 6.00. The Net debit is Rs. 11.45 which is also the maximum possible loss.

STRATEGY : BUY 1 ITM CALL OPTION (LOWER STRIKE), SELL 1 ITM CALL OPTION (LOWER MIDDLE), SELL 1 OTM CALL OPTION (HIGHER MIDDLE), BUY 1 OTM CALL OPTION (HIGHER STRIKE)

Nifty index	Current Value	3600
Buy 1 ITM Call Option	Strike Price (Rs.)	3400
Mr. XYZ pays	Premium (Rs.)	41.25
Sell 1 ITM Call Option	Strike Price (Rs.)	3500
Mr. XYZ receives	Premium (Rs.)	26.00
Sell 1 OTM Call Option	Strike Price (Rs.)	3700
Mr. XYZ receives	Premium (Rs.)	9.80
Buy 1 OTM Call Option	Strike Price (Rs.)	3800
Mr. XYZ pays	Premium (Rs.)	6.00
	Break Even Point (Upper) (Rs.)	3788.55
	Break Even Point (Lower) (Rs.)	3411.45

The Payoff Schedule

On expiry Nifty Closes at	Net Payoff from 1ITM Call purchased (Rs.)	Net Payoff from 1 ITM Call sold (Rs.)	Net Payoff from 1 OTM Call sold (Rs.)	Net Payoff from 1 OTM Call purchased (Rs.)	Net Payoff (Rs.)
3000.00	-41.25	26	9.80	-6	-11.45
3100.00	-41.25	26	9.80	-6	-11.45
3200.00	-41.25	26	9.80	-6	-11.45
3300.00	-41.25	26	9.80	-6	-11.45
3400.00	-41.25	26	9.80	-6	-11.45
3411.45	-29.80	26	9.80	-6	0.00
3500.00	58.75	26	9.80	-6	88.55
3600.00	158.75	-74	9.80	-6	88.55
3700.00	258.75	-174	9.80	-6	88.55
3788.55	347.30	-263	-78.8	-6	0.00
3800.00	358.75	-274	-90.2	-6	-11.45
3900.00	458.75	-374	-190.2	94	-11.45
4000.00	558.75	-474	-290.2	194	-11.45
4100.00	658.75	-574	-390.2	294	-11.45
4200.00	758.75	-674	-490.2	394	-11.45

The payoff chart (Long Call Condor)



SHORT CALL CONDOR

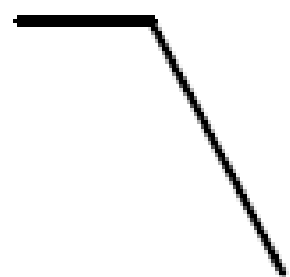
Example: Nifty is at 3600. Mr. XYZ expects high volatility in the Nifty and expects the market to break open significantly on any side. Mr. XYZ sells 1 ITM Nifty Call Options with a strike price of Rs. 3400 at a premium of Rs. 41.25, buys 1 ITM Nifty Call Option with a strike price of Rs. 3500 at a premium of Rs. 26, buys 1 OTM Nifty Call Option with a strike price of Rs. 3700 at a premium of Rs. 9.80 and sells 1 OTM Nifty Call Option with a strike price of Rs. 3800 at a premium of Rs. 6.00. The Net credit is of Rs. 11.45.

STRATEGY : SHORT 1 ITM CALL OPTION (LOWER STRIKE), LONG 1 ITM CALL OPTION (LOWER MIDDLE), LONG 1 OTM CALL OPTION (HIGHER MIDDLE), SHORT 1 OTM CALL OPTION (HIGHER STRIKE)		
Nifty index	Current Value	3600
Sell 1 ITM Call Option	Strike Price (Rs.)	3400
	Premium (Rs.)	41.25
Buy 1 ITM Call Option	Strike Price (Rs.)	3500
	Premium (Rs.)	26.00
Buy 1 OTM Call Option	Strike Price (Rs.)	3700
	Premium (Rs.)	9.80
Sell 1 OTM Call Option	Strike Price (Rs.)	3800
	Premium (Rs.)	6.00
	Break Even Point (Upper) (Rs.)	3788.55
	Break Even Point (Lower) (Rs.)	3411.45

The Payoff Schedule:

On expiry Nifty Closes at	Net Payoff from 1 ITM Call sold (Rs.)	Net Payoff from 1 ITM Call purchased (Rs.)	Net Payoff from 1 OTM Call purchased (Rs.)	Net Payoff from 1 OTM Call sold (Rs.)	Net Payoff (Rs.)
3000.00	41.25	-26	-9.80	6	11.45
3100.00	41.25	-26	-9.80	6	11.45
3200.00	41.25	-26	-9.80	6	11.45
3300.00	41.25	-26	-9.80	6	11.45
3400.00	41.25	-26	-9.80	6	11.45
3411.45	29.80	-26	-9.80	6	0
3500.00	-58.75	-26	-9.80	6	-88.55
3600.00	-158.75	74	-9.80	6	-88.55
3700.00	-258.75	174	-9.80	6	-88.55
3788.55	-347.30	263	78.75	6	0
3800.00	-358.75	274	90.20	6	11.45
3900.00	-458.75	374	190.20	-94	11.45
4000.00	-558.75	474	290.20	-194	11.45
4100.00	-658.75	574	390.20	-294	11.45
4200.00	-758.75	674	490.20	-394	11.45

The payoff chart (Short Call Condor)



Sell Lower

Strike Call

+



Buy middle

strike call

+



Buy middle

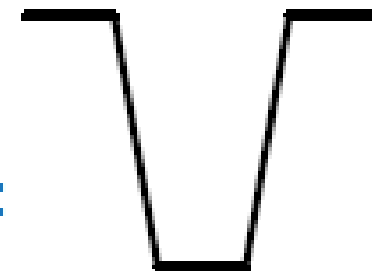
strike call

=



Sell higher

strike call



Short Call

Condor